Anomalous innervations of the extremities are common and influence the interpretation of electrophysiologic studies in normal patients and those with peripheral nerve lesions. The following anomalous innervations are reviewed: median to ulnar nerve communication; ulnar to median nerve communication; variations in the innervation of intrinsic muscles of the hand; accessory deep peroneal nerve; and tibial to peroneal nerve communication. The electrophysiologic recognition of these anomalies and the manner in which they affect the interpretation of electrodiagnostic studies in various conditions is emphasized.
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AAEM MINIMONOGRAPH #2: IMPORTANT ANOMALOUS INNERVATIONS OF THE EXTREMITIES

LUDWIG GUTMANN, MD

Anomalies of peripheral nerves and variations in the innervation of muscles of the extremities and cutaneous areas occur commonly. The anatomy of most of these are reviewed in detail in *Nerves and Nerve Injuries,* by Sydney Sunderland.[33] Knowledge of a few of these is of crucial importance to the clinician–electromyographer in the performance of nerve conduction and electromyographic studies in normal patients and those with peripheral nerve lesions. A lack of awareness of these common anomalies and variations risks faulty interpretation of the electrophysiologic data obtained during nerve conduction and electromyographic examination. This minimonograph will deal with anomalous median-to-ulnar nerve communication and ulnar-to-median nerve communication both in the forearm, variations in the innervations of the intrinsic muscles and sensory dermatomes of the hand, anomalous innervation of the extensor digitorum brevis muscle by the accessory deep peroneal nerve, and an anomalous connection between tibial and peroneal nerves.

From the Department of Neurology, West Virginia University School of Medicine, Morgantown, West Virginia.

Address reprint requests to American Association of Electrodiagnostic Medicine, 21 Second Street S.W., Suite 103, Rochester, MN 55902.

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Send proof and any questions directly to the author: Ludwig Gutmann, M.D., Department of Neurology, West Virginia University Health Sciences Center, Morgantown, WV 26506.

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MEDIAN-TO-ULNAR NERVE COMMUNICATIONS

Communications between the median and ulnar nerves occurring in the forearm have been well described in the literature but receive scant attention in most anatomy books (Fig. 1). The anomaly was described as early as 1763 by the Swedish anatomist R. Martin[23] and later by W. Gruber[10] in 1870. More recent reviews are by Rowntree,[27] Mannerfelt,[20] and Wilbourn and Lambert.[36]

Anatomic Aspects. The median-to-ulnar nerve communications always involve axons descending in the median nerve, crossing through the forearm to join the ulnar nerve, ultimately innervating intrinsic hand muscles. Srinivasan and Rhodes[31] noted the anomaly in 15% of normal fetuses and that it originated from the anterior interosseus nerve in most cases. Less commonly, the anastomosis did not join the ulnar nerve directly but innervated the flexor digitorum superficialis or flexor digitorum profundus. In addition, they noted the anomaly bilaterally in all their cases (8 of 8) of trisomy 21. Although a significant number of axons in a given patient may take this anomalous route, it does not involve all the axons of the median nerve. The median-to-ulnar nerve communications are sometimes referred to as Martin–Gruber anastomosis after its two early describers.

Gruber[10] and Mannerfelt[20] described this anomaly to occur in 15% of extremities studied. Wilbourn and Lambert,[36] using electrophysiologic techniques similar to those described in this monograph, found it to occur in 31% of subjects and 21% of limbs studied. These median nerve axons that cross to the ulnar nerve in the forearm may

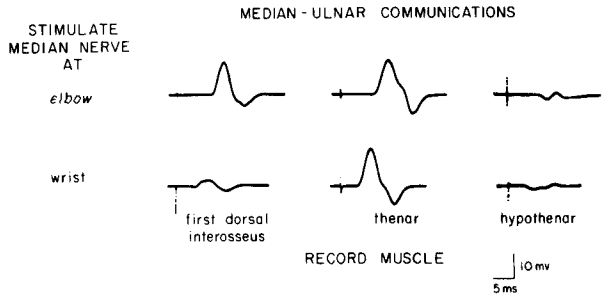


FIGURE 1. CMAP from several muscles on median nerve stimulation at elbow and wrist in a normal individual with median-to-ulnar nerve communications. Stimulation at the elbow evoked a large CMAP from the first dorsal interosseus muscle but not hypothenar. Wrist stimulation evoked a small CMAP from the first dorsal interosseus. The initial positive deflection suggested it arose from thenar muscles. The small hypothenar CMAP preceded by a positive deflection on both elbow and wrist stimulation suggested it was volume conducted from thenar muscles.

innervate various intrinsic muscles of the hand. In a total of 6 cases, Mannerfelt[20] found these axons to innervate the first dorsal interosseus in 5, adductor pollicis in 4, and hypothenar muscle in 2. Wilbourn and Lambert,[36] in 22 limbs showing the anomaly, found these median axons to innervate the first dorsal interosseus area 21 times, hypothenar 9 times, and thenar 3 times. Crutchfield and Gutmann[5] noted the anomaly in 28% of 50 persons and 62% of family members of 5 propositi with this variant. This suggested a dominant mode of inheritance.

Electrophysiology — General Aspects. The anomaly involves motor axons almost exclusively. The studies of Srinivasan and Rhodes[31] would predict this to be the case, because the anastomosis in most cases occurs through the anterior interosseus nerve and this is a pure motor branch of the median nerve. Involvement of sensory axons is unusual, but was reported in 1 case by Santoro et al.[28]

Valls-Sole[34] reports a case of carpal tunnel syndrome and median-to-ulnar communication in whom an additional sensory anomaly occurred. The ulnar side of the third finger and the radial side of the fourth were supplied by ulnar nerve axons that were confined to the ulnar nerve and did not traverse the median nerve at any point. Despite the presence of both a motor and sensory anomaly involving the hand, they were not linked together, again speaking to the expected purely motor component of the median-to-ulnar communication in most cases.

This anomaly may be suspected in otherwise normal individuals when carrying out routine motor conduction studies. In the median nerve, the compound muscle action potential (CMAP) tends to be mildly higher in amplitude (and mildly shorter in duration) when the nerve is stimulated supramaximally distally (wrist) as compared with more proximally (elbow). This is due to increased synchronization at distal stimulation. The amplitude of the CMAP on wrist stimulation is normally larger than that on elbow stimulation and this increase is up to 120%. In the presence of the anomaly, the CMAP on elbow stimulation may be significantly larger. This is so because there are more axons available for stimulation in the median nerve at the elbow than the wrist. Although the CMAP is, in each case, recorded from the thenar muscle group (with the recording electrode usually lying over the motor point of the abductor pollicis brevis), contributions from first dorsal interosseus, adductor pollicis, and thenar muscles innervated by the anomalous axons (located in the ulnar nerve at the wrist) contribute to the larger CMAP on median nerve stimulation at the elbow.

Determining whether this is in fact the case requires supramaximal median nerve stimulation while recording from first dorsal interosseus and hypothenar muscles. If the anomalous axons are present, stimulation of the median nerve at the elbow will evoke a CMAP from one or both muscles. Wrist stimulation of the median nerve will evoke a much smaller or no CMAP from the first dorsal interosseus and no CMAP from the hypothenar muscle.

In Figure 1, from a normal individual, stimulation of the median nerve at the elbow evoked a much larger CMAP from the first dorsal interosseus muscle than stimulation at the wrist. There is no evidence of median crossover innervation to the hypothenar muscle. Figure 2 shows the same occurrence in a patient with concomitant carpal tunnel syndrome with median to ulnar anastomoses involving both first dorsal interosseus and hypothenar muscles.

Conversely, conduction studies in the ulnar nerve may show hypothenar and/or first dorsal interosseus CMAPs that are much larger (greater than 120%) on wrist stimulation than elbow stimulation. This increase is more than that expected from just synchronization and, of course, reflects the increased number of axons present in the ulnar nerve at the wrist as a result of the anomaly (Fig. 3). Such ulnar nerve changes might also indicate an ulnar neuropathy if the concomitant median nerve conduction changes, described above,

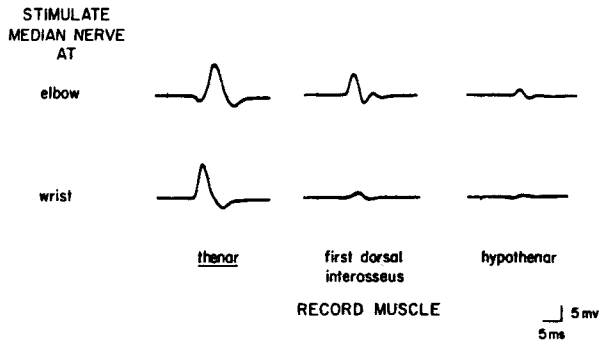


FIGURE 2. CMAP evoked from several muscles on median nerve stimulation at the elbow and wrist in a patient with both CTS and median-to-ulnar nerve communications. The thenar CMAP on elbow stimulation is preceded by a positive deflection not seen on wrist stimulation. Stimulation at the elbow evoked a CMAP from the first dorsal interosseus and hypothenar muscles. Wrist stimulation evoked a very small CMAP which, by virtue of the preceding positive deflection, probably arose from the thenar muscle. The motor conduction velocity (recording thenar muscle) was spuriously fast at 74.4 m/s (difference in latencies is 2.5 ms over a distance of 186 mm).

are not present. It is rather important, at this point, to make certain that the increased size of the CMAP on ulnar stimulation at the wrist is not related to the spread of stimulus to the median nerve lying nearby at the wrist. These changes in the ulnar nerve (i.e., a much larger hypothenar or first dorsal interosseus muscle CMAP on wrist compared with elbow stimulation) may be easily misinterpreted by the unwary electromyographer as a physiologic block (neurapraxia) above the elbow in the ulnar nerve. That these findings are due to the median–ulnar nerve communications is, however, easily documented by demonstrating the smaller CMAP not only on stimulation of the ulnar nerve above the elbow but also immediately below the elbow (arguing against a physiologic block in the region), in addition to seeing the appropriate changes noted above on median nerve stimulation. This is an extremely important point. Last, Kimura[16] has demonstrated that this anomaly can be easily documented using the collision technique.

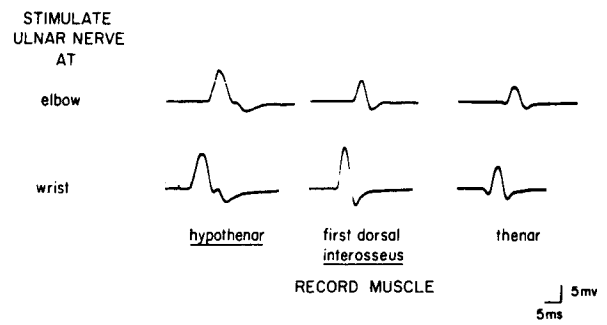


FIGURE 3. CMAP from several muscles on ulnar nerve stimulation at the elbow and the wrist in a patient with CTS and median-to-ulnar nerve communications. First dorsal interosseus CMAP is larger on stimulation at the wrist than elbow. Thenar and hypothenar CMAP are slightly larger on wrist stimulation.

Electrophysiologic Aspects—Median Nerve Lesions. When the anomaly occurs in association with median and ulnar nerve lesions, it may produce additional electrophysiologic changes that make interpretation difficult unless the electromyographer has clear awareness of the anomaly. In the presence of a carpal tunnel syndrome (CTS), median nerve stimulation at the elbow produces a thenar CMAP that is preceded by an initial positive deflection despite the fact that recording electrode lies over the motor point. This initial positive deflection is not seen on median nerve stimulation at the wrist (Figs. 2 and 4). If it is, this now reflects faulty recording electrode placement. Median and ulnar nerve stimulation studies while recording from thenar, hypothenar, and first dorsal interosseus muscles (as described above) will document that this characteristic change in the CTS is due to the anomaly.[8]

Gutmann[8] noted this change in 20% of median nerves in 63 consecutive patients with bilateral CTS. This is in keeping with the incidence of the anomaly. The thenar CMAP preceded by an initial positive deflection on elbow stimulation, but not on wrist stimulation, occurs for the following reason. Conduction in the median nerve axons going through the carpal tunnel and innervating thenar

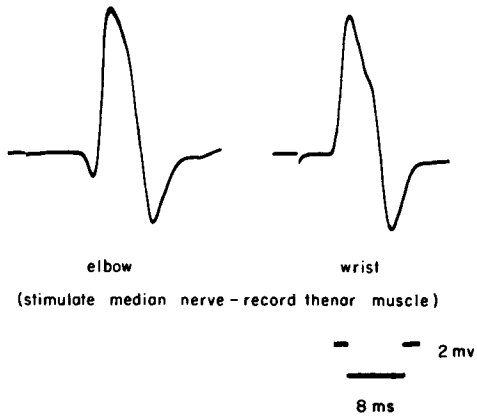


FIGURE 4. Thenar CMAP following median nerve stimulation at the elbow and wrist in a patient with CTS and median-to-ulnar nerve communications. The CMAP evoked on elbow stimulation is slightly larger and preceded by an initial positive deflection. In contrast to Figure 2, median ulnar conduction velocity was normal.

muscles is slower than those median nerve axons crossing to the ulnar nerve and supplying first dorsal interosseus, adductor pollicis, abductor digiti minimi, and flexor pollicis brevis. The CMAP from the latter muscles, on median nerve stimulation at the elbow, is generated prior to that from thenar muscles innervated by median axons going through the carpal tunnel. The earlier appearing CMAP is incorporated into the overall thenar CMAP, but, because it originates at some distance from the recording electrode, produces an initial positive deflection. This initial positive deflection is due to the volume conduction effect and is not present when stimulating the median nerve at the wrist, below the level of the anomalous interchange.

In 6 patients with characteristic symptoms of carpal tunnel syndrome studied by Gutmann et al.,[9] electrophysiologic studies were normal except for the initial positive deflection of the thenar CMAP with median nerve stimulation at the elbow but not at the wrist (Fig. 5). Specifically, distal motor latencies and sensory latencies were normal in median nerves. As normal electrophysiologic studies occur in carpal tunnel syndrome (up to 8%) and median-to-ulnar communications are common

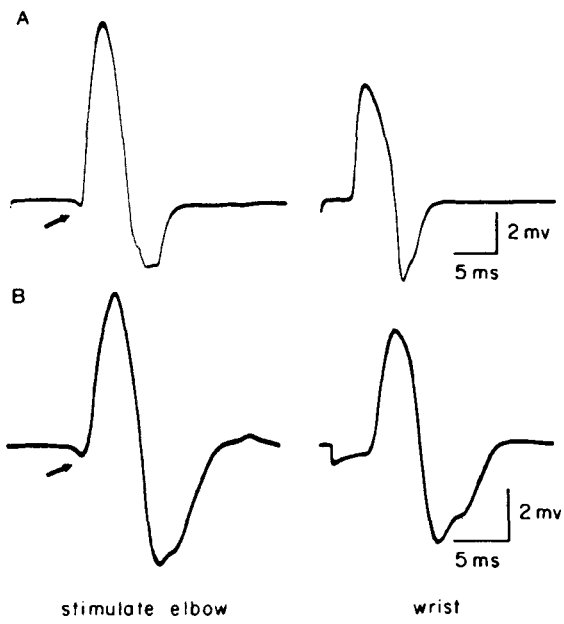


FIGURE 5. Thenar CMAPs evoked by stimulating the median nerve at the elbow and the wrist from 2 cases, **(A)** and **(B)**. Both had symptoms of CTS but normal distal motor and palmar sensory latencies. Arrow denotes initial positive deflection seen on elbow stimulation only. The larger amplitude of thenar CMAP on stimulation at the elbow compared with wrist is also a reflection of the median-to-ulnar communications.

(15–31%), this observed finding can be of practical clinical importance in the diagnosis of mild carpal tunnel syndrome, despite otherwise normal electrophysiologic studies, when the anomaly is present.

It is important to emphasize that when conduction studies are carried out carefully, the occurrence of a thenar CMAP preceded by an initial positive deflection on elbow stimulation but not wrist stimulation is a reliable additional indicator for the presence of a CTS when the median–ulnar nerve anomaly is present. I have not seen this change when conduction studies are carefully performed in patients without a CTS.

Another less common finding noted in CTS in the presence of the anomaly, reported by Kimura[16] and Iyer and Fenichel,[14] is an erroneously normal proximal (elbow) motor latency in the median nerve with prolongation of the distal motor latency. The near normal proximal latency is, again, due to the median nerve communications (for the reasons noted above) and results in a spuriously fast-calculated conduction velocity. In the 63 consecutive patients studied by Gutmann[8] this finding occurred only once (Fig. 2).

Additionally, Lambert[18] described a thenar CMAP with 2 components on median nerve stimulation at the elbow, again related to the slower conduction in the median nerve at the wrist and faster conduction in those axons that have crossed to the ulnar nerve. The ulnar-nerve component is absent when the median nerve is stimulated at the wrist (Fig. 6). This too appears to be an uncommon finding.

Electrophysiologic Aspects—Ulnar Nerve Lesions. This anomaly may also affect the interpretation of motor conduction studies, as well as the clinical examination, in the presence of an ulnar-nerve lesion. This is best exemplified by a complete ulnar neuropathy due to a lesion at the level of the elbow. In this circumstance, stimulation of the ulnar nerve just above and below the elbow evokes no hypothenar or first dorsal interosseus CMAP while stimulation at the wrist does. The same changes could, of course, be produced by a physiologic block of the ulnar nerve in the forearm. That this change is due to the median—ulnar communications is further demonstrated by stimulation of the median nerve. The hypothenar and first dorsal interosseus CMAP is evoked on elbow stimulation while wrist stimulation evokes little or no response (Fig. 7). This clearly documents that the hypothenar and first dorsal interosseus CMAPs are derived

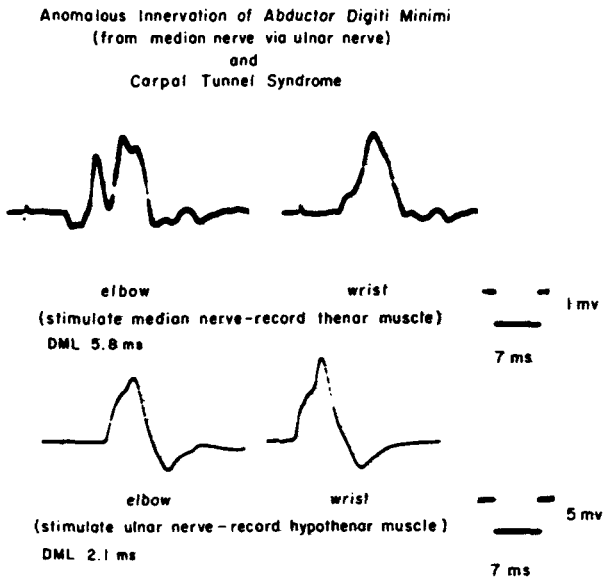


FIGURE 6. Conduction studies in a patient with CTS and median-to-ulnar nerve communications. Stimulation of median nerve at elbow evokes a thenar CMAP with both two components and an initial positive deflection. The first component is due to the faster conduction in those axons crossing to the ulnar nerve (this also results in spuriously fast proximal latency and conduction velocity). Neither change is noted on wrist stimulation of median nerve. Ulnar nerve conduction studies show a larger hypothenar CMAP on wrist stimulation.

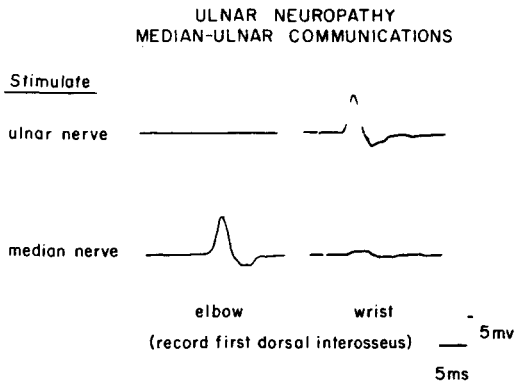


FIGURE 7. Ulnar and median nerve conduction studies in a patient with complete transection of the ulnar nerve at the elbow and median-to-ulnar nerve communications. A first dorsal interosseus CMAP is evoked on ulnar nerve stimulation at the wrist, but not the elbow. The same first dorsal interosseus CMAP is evoked on median nerve stimulation at the elbow, but not the wrist.

from stimulated axons lying in the ulnar nerve at the wrist and median nerve at the elbow. A correct interpretation of these findings would avoid the incorrect conclusion that some ulnar nerve axons going through the lesion at the elbow are still anatomically intact.

In a case reported by Miller,[24] the anomaly complicated the evaluation of an ulnar neuropathy localized to the cubital tunnel. This resulted in the ulnar conduction block appearing more severe by 2 mV (stimulating the median nerve at the elbow resulted in a 2-mV CMAP from the abductor digiti minimi). Nonetheless, even after subtracting this 2-mV contribution, the amplitude reduction in the ulnar nerve stimulating above the cubital tunnel was still approximately 50%.

This anomaly may complicate the clinical examination when an ulnar neuropathy with severe or complete axonal loss occurs. Because of median-to-ulnar communicating axons innervating hypothenar, first dorsal interosseus, and adductor pollicis muscles, the degree of clinical weakness may be considerably less than anticipated. When the median-to-ulnar nerve communications in the forearm are especially generous, there may be only mild weakness of intrinsic muscles of the hand despite complete degeneration of the ulnar nerve at the elbow. It is under this circumstance that the "all median hand" occurs as described by Marinacci[21]; however, the "all median hand" occurs only in the presence of an ulnar neuropathy in our experience. At best, one can only see a "too much median hand" in the presence with median-to-ulnar nerve communications in persons having no nerve lesions.

ULNAR-TO-MEDIAN NERVE COMMUNICATIONS

In contrast to the common occurrence of median-to-ulnar communications, that of ulnar-to-median communications is rare. They have been reported on three occasions involving motor axons and once involving only sensory axons.

The initial case of Marinacci[22] involved a patient with median nerve trauma at the forearm. Denervation occurred in the forearm flexor muscles, usually supplied by the median nerve, but hand muscles remained unaffected. Motor axons supplying the thenar muscles could be stimulated in the median nerve at the wrist or ulnar nerve at the elbow. A case described by Komar and colleagues[17] had only partial loss of thenar function and hand sensation after a complete median nerve transection above the elbow. A 15-mV thenar CMAP was evoked on ulnar stimulation at the elbow but not the wrist. Streib[32] described a woman with fatigue and generalized weakness showing the thenar CMAP 4.9 mV larger stimulating the median nerve at the wrist than elbow and 6.6 mV larger stimulating ulnar nerve at the elbow than wrist. No evidence of sensory axon abnormality was noted. The findings in these cases clearly doc-

ument axons to thenar muscles in the ulnar nerve at the elbow that crossover to the median nerve in the forearm in the median nerve at the wrist.

Hopf[12] noted ulnar-to-median communication limited to sensory axons in 1 of 30 normal subjects in a study of ulnar sensory distribution. The sensory nerve action potentials evoked by stimulating the digital nerves of the ulnar side of the middle finger and radial side of the ring finger were recorded from the median nerve at the wrist and ulnar nerve at the elbow. Motor conduction studies showed no similar communication.

Although uncommon, an awareness of this anomaly is useful. It can effect the clinical and electrophysiologic presentation of ulnar and median neuropathies.

VARIATIONS IN THE INNERVATIONS OF INTRINSIC MUSCLES OF THE HAND

All the intrinsic muscles of the hand are innervated by the ulnar and median nerves but many variations exist. This is in part due to the median–ulnar nerve communications discussed in the previous section. These variations of muscle innervation, however, may occur without actual anomalous communications between these two nerves in the forearm or the hand.

Anomalous communications in the hand were described by Riche[26] and Cannieu[2] and are sometimes referred to as the Riche–Cannieu anastomosis. They are not as well defined as the median–ulnar nerve communications in the forearm. They are thought to involve communications between the motor branch of the median nerve and the deep branch of the ulnar nerve in the radial part of the hand. The anatomic and clinical importance of these Riche–Cannieu anastomoses has not been clarified. It is not clear whether these communications are sensory, motor, or mixed, and if they pass from ulnar to median nerve or in the opposite direction. Mannerfelt[20] noted these communications in 3 of 9 hand dissections. Harness and Sekeles[11] noted anastomoses between the deep branch of the ulnar nerve and various branches of the median nerve deep to the thenar muscles in 27 of 35 (77%) hand dissections.

Most commonly, the first dorsal interosseus, adductor pollicis, and abductor digiti minimi are innervated by the ulnar nerve and the abductor pollicis brevis and opponens pollicis by the median nerve. The flexor pollicis brevis may have a dual nerve supply with the median nerve innervating the superficial head and the ulnar nerve the deep head.

Rowntree's[27] study (recently again reviewed by Seddon[29]) documented many variations in the pattern of innervation in 226 cases with ulnar and median nerve lesions: 33% of patients showed all thenar muscles (flexor pollicis brevis, abductor pollicis brevis, and opponens pollicis) to be entirely innervated by the median nerve; 32% showed median nerve innervation of the abductor pollicis brevis and opponens pollicis and the entire flexor pollicis brevis by the ulnar nerve; 15% showed innervation of the abductor pollicis brevis and opponens pollicis by the median nerve and dual innervation of the flexor pollicis brevis; 2% of the cases were interpreted as having all intrinsic hand muscles supplied by the ulnar nerve; and 1%, the entire thenar musculature and adductor pollicis were interpreted as being exclusively supplied by the median nerve while the first dorsal interosseus and abductor digiti minimi were supplied by ulnar nerve. In our laboratory, however, we have not seen examples of all intrinsic hand muscles innervated entirely by the ulnar nerve or the first dorsal interosseus and adductor pollicis innervated entirely by the median nerve. Rowntree apparently recognized that the hypothenar muscles may at times have innervation from the median nerve but did not comment on this.

Kimura and Ayyar[15] investigated the median and ulnar communications in the hand stimulating the median and ulnar nerves at the wrist recording from abductor pollicis brevis, first dorsal interosseus, and abductor pollicis brevis muscles using surface electrodes. Ulnar nerve innervation to thenar muscles was 83%. Median nerve innervation to first dorsal interosseus was 43% and abductor digiti minimi 16%. The changes were more common in caucasians and hispanics than blacks. Although the numbers of communication may be inflated by virtue of the volume conduction effect, the findings again indicate the common occurrence of these communications.

Two recent publications highlight the clinical importance of these variations. Dumitru et al.[6] documented a patient with an ulnar neuropathy at the elbow in whom denervation seen in the abductor pollicis brevis was due to an anastomosis from the deep ulnar nerve in the hand to partially innervate the abductor pollicis brevis. Seradge and Seradge[30] documented 2 patients with carpal tunnel syndromes showing partial and complete innervation, respectively, of the abductor digiti minimi by a nerve branch leaving the median nerve at the midcarpal tunnel area. It then crossed the tunnel superficial to the flexor tendons and pene-

trated the transverse carpal ligament to innervate the muscle.

An awareness of these variations is essential for the correct interpretation of the findings on clinical examination and needle electromyogram. Especially important is that signs of denervation may occur in flexor pollicis brevis with ulnar nerve lesions because of its frequent ulnar innervation. Further, one may see denervation in first dorsal interosseus with median lesions and abductor pollicis brevis with ulnar nerve lesions on rare occasions due to anomalies cited above.

ACCESSORY DEEP PERONEAL NERVE

The clinical importance of the extensor digitorum brevis muscle (EDB) is that it extends the second to fifth toes and is commonly used to record CMAPs when carrying out peroneal nerve conduction studies. The EDB is usually innervated exclusively by the deep peroneal nerve, a major branch of the common peroneal nerve; however, in as many as 28% of patients, 1 or both of the EDB muscles are partially innervated by the accessory deep peroneal nerve, a branch of the superficial peroneal nerve. Anatomic studies by Bryce,[7] Winckler,[37] and Crutchfield[3] have demonstrated that the accessory deep peroneal nerve may innervate the lateral portion of the EDB. The accessory deep peroneal nerve arises from the superficial peroneal nerve midday on the lateral aspect of the leg. It passes deep and posterior to the peroneus brevis tendon, behind the lateral malleolus, and subsequently enters the lateral portions of the EDB (Fig. 8). Electrophysiologic studies have demonstrated this anomaly, involving one or both EDB muscles, in 21–28% of individuals.[3,19,37] This anomaly appears to be inherited in a dominant fashion. Crutchfield and Gutmann[4] showed that 78% of relatives of 5 subjects also had the anomaly, compared with 22% of unrelated individuals.

When carrying out peroneal nerve conduction studies, the amplitude of the EDB CMAP, stimulating the deep peroneal nerve at the ankle, should be 90–120% of the amplitude compared with stimulating the common peroneal nerve at the knee. In the presence of an accessory deep peroneal nerve innervating the EDB, the EDB CMAP is considerably smaller when stimulating the deep peroneal nerve at the ankle because some of the axons of the common peroneal nerve to the EDB are in the accessory deep peroneal nerve. This can be proven by stimulating the accessory deep peroneal nerve behind the lateral malleolus, evoking an additional CMAP from the EDB (Fig. 9). Although this anomaly does not mimic any pathologic lesion, the smaller CMAP on supramaximal stimulation of the deep peroneal nerve at the ankle may be difficult for the electromyographer to explain unless he is aware of the anomaly.

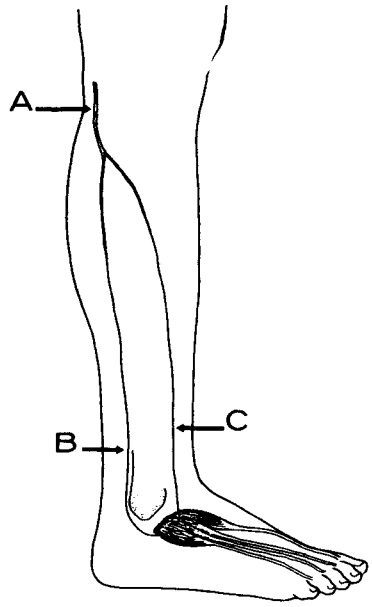


FIGURE 8. Diagrammatic representation of **(A)** common peroneal nerve, **(B)** accessory deep peroneal nerve (branch of superficial peroneal nerve), and **(C)** deep peroneal nerve.

This anomaly may cause minor confusion in

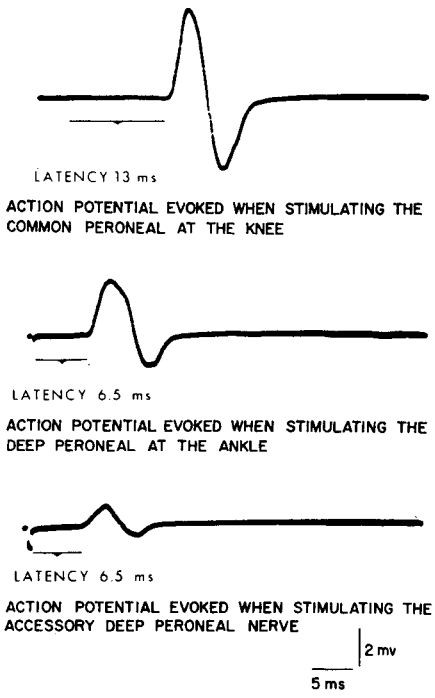


FIGURE 9. CMAPs evoked from EDB in the presence of an accessory deep peroneal nerve.

the evaluation of lesions of the common peroneal nerve. In a partial lesion involving a physiologic block at the knee, the expected smaller CMAP, stimulating above the head of the fibula (compared with at the ankle), may be lost due to the concomitant smaller CMAP on ankle stimulation of the deep peroneal nerve resulting from the anomaly. However, in this circumstance one would expect a larger EDB CMAP on common peroneal nerve stimulation below the head of the fibula than that seen at the other two stimulation sites.

The accessory deep peroneal nerve may cause considerable difficulty in both the clinical and electrophysiologic evaluation of deep peroneal neuropathies.[7] These latter, of course, occur much less commonly than neuropathies of the common peroneal nerve. In a neuropathy of the deep peroneal nerve, as due to nerve compression by osteochondroma or ganglion cyst, one would expect weakness of the anterior tibialis and EDB muscles with complete sparing of muscles innervated by the superficial peroneal nerve (i.e., peroneus longus and brevis). In the presence of an accessory deep peroneal nerve, there would be unexpectedly good function of the EDB (especially in its lateral portion innervating the fourth and fifth digits). A complete or almost complete neuropathy of the deep peroneal nerve showing complete or almost complete loss of function in the anterior tibialis muscle and good function in the EDB might lead to confusing and erroneous conclusions.

Similarly, stimulation of the deep peroneal nerve at the ankle, in this circumstance, would evoke a very small or no EDB CMAP while stimulation of the common peroneal nerve at the knee would evoke a much larger (and possibly normal) CMAP. The situation would, of course, be clarified once it was discovered that this same CMAP could be evoked on stimulation behind the lateral malleolus, i.e., stimulating the accessory deep peroneal nerve (Fig. 10). From a clinical point of view, residual function in the lateral portion of the EDB muscle, in the face of an otherwise complete deep peroneal palsy, should suggest this anatomic variation.

TIBIAL-TO-PERONEAL NERVE COMMUNICATION

Phillips and Morgan[25] recently described the findings in a case of tibial–peroneal nerve communication on the basis of intraoperative nerve conduction studies. Stimulation of the tibial nerve produced a contraction from the peroneus longus muscle and a nerve action potential in the distal peroneal nerve. This crossover resulted in confusing clinical and electrophysiologic findings including the presence of a few motor unit potentials in the peroneus longus muscle with otherwise complete peroneal denervation. A prior tibial-peroneal communication was reported but no details given.[35]

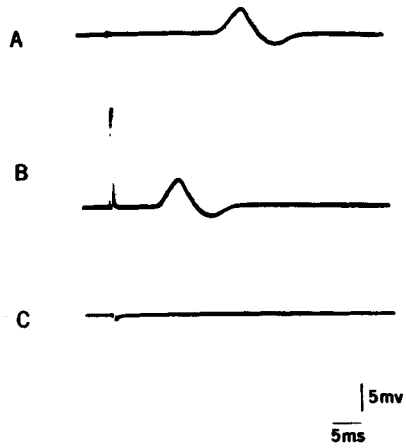


FIGURE 10. A case of complete deep peroneal neuropathy in the presence of an intact accessory deep peroneal nerve. CMAPs evoked from EDB on stimulation of **(A)** common peroneal nerve, **(B)** accessory deep peroneal nerve, and **(C)** deep peroneal nerve. No CMAP was evoked on stimulating (C).

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